

Tutorial 01 - Number Systems

1. Define the following terms.
 - a. Least significant bit
 - b. Most significant bit

2. How many digits are used in the binary number system?

3. Convert these decimal numbers into binary using the Division Algorithm.

a. 257	d. 1040
b. 32	e. 576
c. 790	

4. Convert these Binary numbers into decimal using the Multiplication Algorithm.

a. 10011	e. 10001111
b. 1011	f. 101010
c. 1010	g. 100111
d. 110000	

5. Convert these decimal numbers to hexadecimal numbers.

a. 392	e. 834
b. 24	f. 52
c. 734	g. 46
d. 267	

6. Convert these hexadecimal numbers into decimal numbers

a. 34E	e. BFA2
b. 163F	f. 4CD9
c. 6F7D	g. 72E
d. FB	

7. Convert these binary numbers into hexadecimal numbers using the 4-bit substitution method.

a. 11010101110101110011111

b. 101010001100101010101

c. 10100011101010101

d. 001010101000110101

e. 1010001010101001

f. 010101011010101001111

g. 1111001010101111111

h. 101000101010101010

Substitution Code			
0000 = 0	0100 = 4	1000 = 8	1100 = C
0001 = 1	0101 = 5	1001 = 9	1101 = D
0010 = 2	0110 = 6	1010 = A	1110 = E
0011 = 3	0111 = 7	1011 = B	1111 = F